PERMUTATION & COMBINATION:
(क्रमचय और संचय):

PERMUTATION & COMBINATION formulas:

Permutation

Select & arrange

$= {}^{n}C_r \times r!$

$= {}^{n}P_r$

$= \frac{{}^{n}P_r}{r!}$

$= {}^{n}C_r$

Combination

Selection

${}^{n}C_r$

$^{10}P_2 = 10 \times 9$

$^{10}P_3 = 10 \times 9 \times 8$

$^{7}P_2 = 7 \times 6$

$^{34}P_5 = 34 \times 33 \times 32 \times 31 \times 30$

!/L ⇒ 1 से 8

$^{10}C_2 = \frac{10 \times 9}{2!}$

$^{10}C_3 = \frac{10 \times 9 \times 8}{3!}$

$^{7}C_2 = \frac{7 \times 6}{2!}$

$^{34}C_5 = \frac{34 \times 33 \times 32 \times 31 \times 30}{5!}$

- 5 boys are to be arranged on 7 chairs
- 5 students are to be selected from 10 students

- 5
- 5

In how many ways can we select 2 persons from a group of 20 persons such that

1) they are to be chosen for a project work
2) they are to be selected for posts of president and vice president
3) they are to be chosen for posts of directors

1)

2)

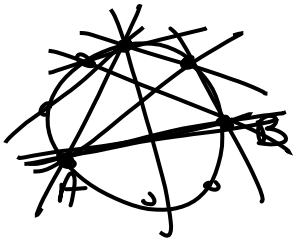
3)

In how many ways can we gift 3 cars to 3 persons among a group of 20 persons such that

1) all the cars are of the same type
2) all the cars are of different type

1)
2)

- Number of handshakes between 10 persons
- Total number of lines formed by joining 8 points on a circle



Important points:

1. The number of permutations of n different objects taken r at a time, where $0 \leq r \leq n$ and the objects do not repeat is n_{p_r}.
2. 'n' objects can be arranged in a row in n! ways.
3. 'n' objects out of which 'r' are identical, can be arranged in a row in $\frac{n!}{r!}$ ways.
4. 'n' objects can be arranged in a circle in $(n-1)!$ ways.
5. 'r' objects can be selected out of 'n' objects where $0 \leq r \leq n$ in n_{C_r} ways.

10

1st 2nd 3rd

$10 \times 9 \times 8 \times \ldots \times 1 = 10!$

TAANA

$\frac{5!}{3!}$

$T A_1 A_2 N A_3$

$T A_1 A_3 N A_2$

$T A_2 A_1 N A_3$

$T A_2 A_3 N A_1$

$3! = 6$

STAR

$4! = 24$ ways

ART.

ATR.

RAT.

RTA.

TAR

TRA

$3! = 6$ ways

MISSISSIPPI

$$\frac{11!}{4!\,4!\,2!} \text{ ways}$$

6 person

1st 2nd 3rd

6 5 4 . . . x 1 = 6!

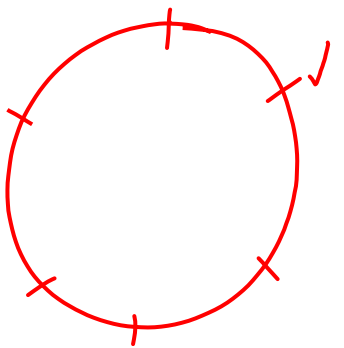
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Dessert

6 person

1st 2nd

1 x 5 x 4 x 3 x 2 x 1 = 5!



40. **In how many ways 7 boys can sit in a row?**

a) 5040 b) 720

c) 28 d) 120

41. **In how many ways 7 boys can sit in a circle?**

a) 5040 b) 720

c) 21 d) 120

42. Rahul invited his 19 friends to his birthday party. They all gathered to have dinner around a circular table. In how many ways they can sit such that Rahul wants to sit between Amit and Sumit?

a) $19!$
b) $18!$
c) $2 \times 18!$
d) $2 \times 17!$

43. Rahul invited his 19 friends to his birthday party. They all gathered to have dinner around a circular table. In how many ways they can sit such that Salman and Vivek should not sit together?

a) $340 \times 18!$
b) $340 \times 17!$
c) $17 \times 18!$
d) $18 \times 17!$

44. Ten chairs are placed in a row which have to be occupied by 8 students and two teachers. The two teachers decide not to sit together. If there are n such arrangements of occupying the seats, then what is the sum of digits of n?

a) 18 b) 19 c) 17 d) 16

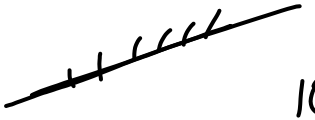
PnC (row)

45. In how many ways can we make garland from 5 different flowers?

a) 12
b) 24
c) 120
d) 60

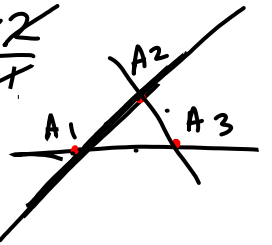
Misc points:

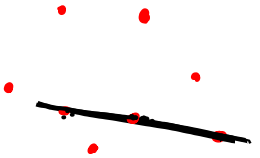
1. The maximum number of lines that can be drawn using ‘n’ non collinear points is n_{c_2}.
2. The maximum number of lines that can be drawn using ‘n’ points out of which ‘r’ points are collinear is $n_{c_2} - r_{c_2} + 1$
3. The maximum number of triangles that can be formed using ‘n’ non collinear points is n_{c_3}.
4. The maximum number of triangles that can be formed using ‘n’ points out of which ‘r’ points are collinear is $n_{c_3} - r_{c_3}$
5. The maximum number of intersection points formed by drawing ‘n’ lines is n_{c_2}.
6. The maximum number of intersection points formed by drawing ‘n’ circles is $2 \times n_{c_2}$.



$10C_2 = \frac{\cancel{10}^{5} \times 9}{\cancel{2!}} = 45$

$^{3}C_2 = \frac{3 \times \cancel{2}}{\cancel{2!}}$



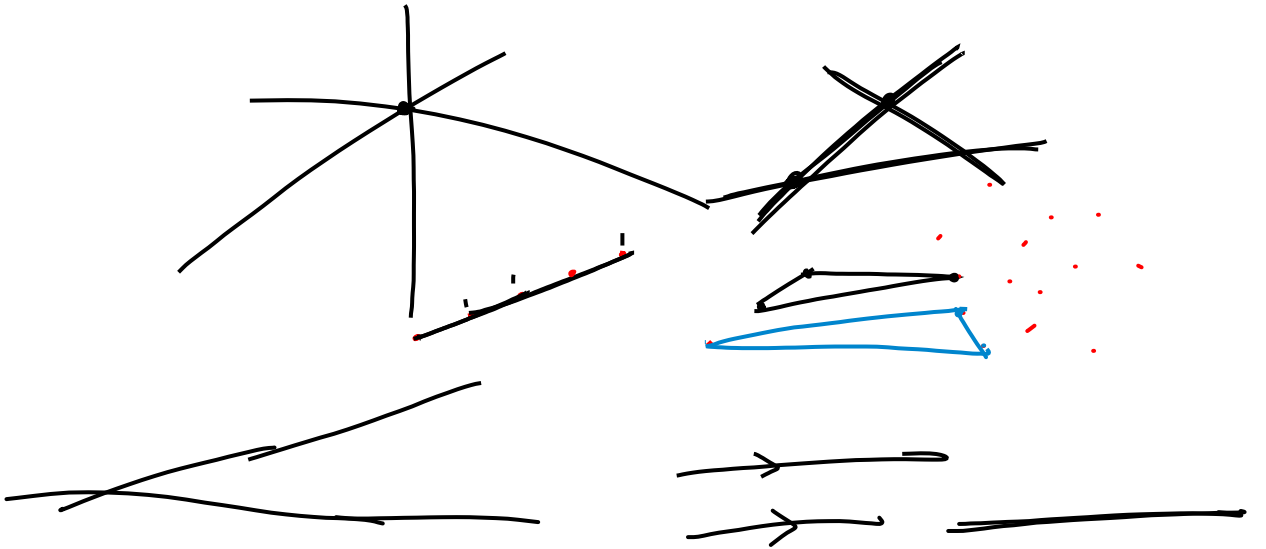


$^{8}C_2 - {}^{3}C_2 + 1$

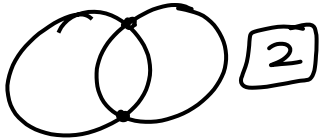


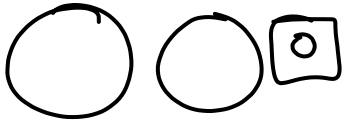
8 (3 collinear)

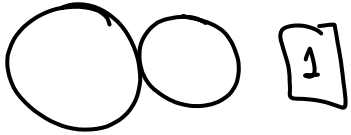




$n C_2 \times 2$

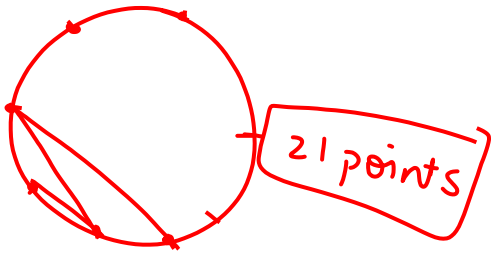






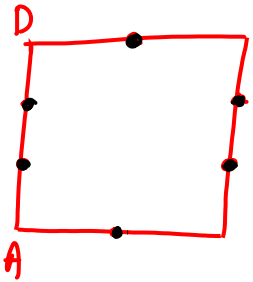
46. **How many chords can be drawn through 21 points on a circle?**

a) 210 **b) 240**
c) 105 **d) 320**



47. $ABCD$ is a square. One point on each of AB and CD; and two distinct points on each of BC and DA are chosen. How many distinct triangles can be drawn using any three points as vertices out of these six points?

a) 16 b) 18
c) 20 d) 24



48. There is a polygon of 11 side. How many triangles can be drawn by only using the vertices of the polygon?

a) 165
b) 150
c) 175
d) 180

SSC CHSL, 8 Jul 2019 S2